Shahyad Khamnei

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Education

Boston University, Boston, MA *Expected December 2027*
B.A. in Mathematics, Physics, & Computer Science
Graduate Coursework: Deep Learning Theory, Probability Theory, Optimization Algorithms, Quantum Computing

University of Geneva, Geneva, Switzerland *February – June 2026*
GPA: 5.75/6.00
Spring 2026 Semester Exchange Student
Graduate Coursework: Measure Theory, Representation Theory & Lie Algebras

Research Experience

Max Planck Institute for the Physics of Complex Systems (MPIPKS), Dresden, Germany *Fall 2026*
Research Intern in Condensed Matter Divison –Nonequilibrium Quantum Dynamics (Advisor: Dr. Marin Bukov)
Project Title: TBD

CERN Theory Division, Geneva, Switzerland *Spring - Summer 2026*
Research Intern (Advisor: Dr. Kyriakos Papadodimas)
Project Title: Random Matrix Theory and Free Probability in Quantum Systems

Boston University Math & Statistics Department, Boston, MA *Fall 2025 – Fall 2026*
Undergraduate Researcher (Advisor: Prof. Konstantinos Spiliopoulos)
Project Title: Algorithms and Optimization of Random Vector Functional Link Neural Networks

- In collaboration with Los Alamos National Lab Applied Computer Science Group.

Boston University Condensed Matter Theory Group, Boston, MA *Summer - Fall 2025*
Undergraduate Researcher (Advisor: Prof. Anatoli Polkovnikov)
Project Title: Partial Reversibility and Counterdiabatic Driving in Nearly Integrable Systems

- Gave a talk presenting our results at American Physical Society 2026 Global Conference. Link Here

University of South Carolina REU, Columbia, SC *Summer 2024*
REU Participant (Group Advisor: Prof. Zhu Wang)
Project Title: Data-Driven Reduced Order Modeling of PDEs

Publications and Preprints

"Partial Reversibility and Counterdiabatic Driving in Nearly Integrable Systems"
Rohan Banerjee, Shahyad Khamnei, Anatoli Polkovnikov, and Stewart Morawetz.
Physical Review A, 2026.

Awards and Recognitions

Boston University Quantum Convergence Grant

- Used to fund my travel to the APS Global Summit 2026

Iranian-American Scholarship Foundation Scholar
Boston University Statistics Department Junior Award for Academic Excellence
Boston University Dean's List

North Charleston West Corridor Merit Scholar

Technical Skills

Technical Skills: Python (PyTorch, Pandas, NumPy, SymPy, MatPlotLib), C++ (Eigen), Java, Mathematica, Linux, Git Hub, Latex
Languages: English (Native Speaker), Farsi (Native Speaker)

Coursework

Math & Statistics: Mathematical Theory of Deep Learning, Probability Theory, Measure Theory, Representation Theory & Lie Algebras, Financial Derivatives, Real Analysis I & II, Complex Analysis, Abstract Algebra I & II, Differential Equations, Linear Algebra, Multivariate Calculus
Physics: Quantum Physics I & II, Quantum Computing, Electricty & Magnetism, Computational Physics
Computer Science: Optimization Algorithms, Algorithms Analysis, Data Structures
Spring 2027: Functional Analysis, Topology, Computer Architecture, TBD

Leadership & University Engagement

Boston University PRISM Mentorship Program *2025 – 2026*
Student Advisor

- Advise first and second year students in math and physics regarding coursework, research, and their career paths.

Boston University Technology Inspiration Scholars Program *2024 – 2025*
Logistics Co-Director

- Organized a hackathon for underprivileged high school students in the Greater Boston area.

Boston University F.E.A.S.T. Branch of Student Government *2023 – 2025*
Treasurer

- F.E.A.S.T. stands for Food Equality, Accountability, Sustainability, and Transparency.
- Organized and Handled Funding for community service events and promoted food sustainability in the community and university.